RISK MANAGEMENT

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Valid until:	05/07 /2028

DOCUMENT HISTORY			
VERSION	CAUSES OF MODIFICATION	EFFECTIVE DATE	REPLACES (Code, Revision)
01	Initial edition	07/02/2027	N/A
02	Changes to Annexes I and II for more information about risks as indicated in AC-044/24, update of annex templates, update of Spika departments	10/05/2024	PNT-002.01
03	Updated the document annexes to include the template for the associated records	June 14, 2024	PNT-002.02
04	Update document to comply with ISO 14971	05/07 /2025	PNT-002.03

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1. Aim

The objective of this procedure is to describe the guidelines to follow for risk management at Spika Tech as a component of the Quality Management System, so that quality risks to manufactured products can be detected and minimized.

2. Scope

This procedure applies to all products manufactured by SPIKA TECH in their design, production, storage, and distribution phases, i.e., the production environment and the processes and subprocesses surrounding it.

The same will apply to all internal SPIKA TECH staff and external personnel who, by decision of the Technical Manager, are interested in adopting the same risk management system.

3. Responsibilities and Departments Affected

SPIKA TECH personnel are responsible for quality risk management and following the instructions in this procedure. All personnel who need to be familiar with this procedure should report any questions, changes to the procedure, or errors to their immediate supervisor, who will then inform the person responsible for developing this procedure so that they can take appropriate action.

Production department managers must conduct a risk assessment whenever there is a change in processes, facilities, equipment, or services, or when new products are added or changes are made to existing ones that could affect the quality of the products being manufactured.

The responsible technician must:

- Participate in the review and management of risk analysis as indicated in this procedure.
- Review and approve the risk analysis report.
- Maintain and archive all department risk analysis reports.
- Maintain current documentation and ensure that it is prepared, distributed, and stored in accordance with established regulations.
- Ensure compliance with the provisions of this procedure.

The affected departments are the following:

Department	Apply

	Yeah	No
Address	☒	☐
Quality Assurance	☒	☐
Logistics and warehouse	☒	☐
Investigation	☒	☐
Development	☒	☐
Marketing and communication	☒	☐
ITEM	☒	☐
Robotics	☒	☐
Clinical studies	☒	☐

4. Definitions and Acronyms

- Risk: combination of the probability of occurrence of a damage and the severity of that damage.
- Risk management: systematic application of management policies, procedures and practices to the tasks of risk analysis, evaluation, control and monitoring.
- Injury: physical harm, damage to health, property or the environment.
- Harm : harm to health, including that which may result from loss of quality or availability of the product .
- Hazard: potential source of harm.
- Hazardous situation: circumstance in which people, property or the environment are exposed to one or more hazards (ISO 14971).
- Adverse event: A product defect (i.e., a malfunction or failure, deterioration of characteristics or performance, or inadequacy of labeling or instructions for use) that,

directly or indirectly, has caused or could have caused serious consequences, such as a deterioration in the health of the patient, user, or other person. Also called an incident.

- Severity: A measure of the potential consequences of a hazard.
- Probability of occurrence: classification of the probability of a failure occurring during the design, manufacturing and marketing of the product.
- Risk assessment: a set of operations that allows for the assessment of potential damage (harm) or injury.
- Risk analysis: a study whose objective is to determine the possible risks that may occur during the operational phase of a product.
- Risk control: The application of a set of operations that allow the evaluation, control, and monitoring of risks associated with a specific product or process, in order to ensure that the identified risks do not occur or have a very low probability.
- Residual risk: risk that remains after applying control measures.
- Risk Management Team (RMT): Representatives from different functions who contribute to the risk assessment based on their specific expertise.
- Corrective action: Action taken to eliminate the causes of a nonconformity, defect, or other undesirable situation, in order to prevent its recurrence.
- Preventive action: Action taken to eliminate the causes of potential nonconformities, defects or other undesirable situations, in order to prevent their occurrence.

5. Procedure

5.1. Generalities

The risk analysis aims to identify the risks associated with the processes applied to the production and control of products manufactured at the SPIKA TECH plant, including medical devices.

Once risks are identified, they can be controlled through appropriate actions focused on minimizing their occurrence and, if they do occur, reducing their consequences.

Effective risk analysis management can ensure the quality of manufactured products by proactively providing measures to identify and control critical quality aspects during production phases. Furthermore, such management improves decision-making when quality issues are detected.

- The purpose of risk management is:
- Integrate the functions of the departments and the company in the process
- Innovate
- Continuously improve processes
- Develop processes based on risk assessment and control

5.2. Method: Tools

The following are the phases for risk management:

5.2.1. Risk management planning

At the start of a project, a Risk Management Plan must be developed. The plan should define, at a minimum:

- Scope of risk management activities
- Risk acceptability criteria
- Roles and responsibilities
- Risk analysis and assessment methods
- Review and update criteria.

5.2.2. Risk assessment

Risk assessment is the fundamental step in the risk management process. It involves systematically identifying hazards, analyzing potential risks, and evaluating their impact on product quality, safety, and regulatory compliance.

If a process, change, or activity could affect product quality or regulatory compliance, a formal risk assessment must be conducted.

a) Hazard Identification : Hazard identification is the foundation of effective risk management. A cross-functional project team with sufficient technical and process knowledge should:
   - Analyze the process or system in detail
   - Identify all potential hazards or sources of damage, including misuse, malfunction, and abnormal conditions
   - Document hazards using appropriate tools such as process maps, flowcharts, or brainstorming sessions.

   Sources of danger can be:

   - Materials and components
   - Device design and software
   - Equipment and manufacturing methods
   - User interactions
   - Environmental factors

b) Analysis : Risk analysis involves estimating the probability of occurrence and the severity of the resulting harm for each identified hazard. SPIKA TECH can select the most appropriate analytical method based on the complexity and context of the process. Common methods include (adapted from ISO 14971 and ICH Q9):
   - Basic methods that allow you to structure risk management using simple tools such as flowcharts, checklists, process diagrams, cause-effect diagrams, etc.
   - Failure Mode and Effects Analysis (FMEA)
   - Risk Matrix (RM)
   - Hazard Analysis and Critical Control Points (HACCP)

- Functional Operability Analysis (FOA)
- Preliminary Risk Analysis (PRA)

The selected method must be documented and justified based on the intended use and risk profile of the process or product.

c) Assessment : Risk assessment involves comparing estimated risks with predefined acceptance criteria to determine their criticality and whether further control is needed. Each risk will be assessed based on:

- Probability (P): expected frequency of occurrence of the identified risk.

Probability of damage	Description	Punctuation
Unlikely	≥ 1 in 1,000,000 (<10−6)	1
Remote	≥ 1 in 100,000 (<10−5 and ≥10−6)	2
Occasional	≥ 1 in 10,000 (<10−4 and ≥10−5)	3
Likely	≥ 1 in 1000 (<10−3 and ≥10−4)	4
Frequent	≥ 1 en 100 (≥10−3)	5

- Severity (S): severity of the occurrence of the identified risk, evaluates the level of consequences.

Severity of damage	Description	Punctuation
Negligible	Causes temporary discomfort or inconvenience	1
Minor	Causes a temporary injury or impairment that does not require medical or surgical intervention . Causes temporary discomfort or inconvenience.	2
Serious / Major	Causes injuries or impairments that require medical or surgical intervention.	3
Critical	Causes permanent disability or irreversible injury	4
Catastrophic / Fatal	Causes death	5

5.2.3. Risk acceptance

Once each risk has been assessed, the RPN (Risk Priority Number) is obtained by applying the following formula:

$$RPN = P \times S$$

Each estimated risk must be systematically evaluated, keeping in mind that risks should be reduced to the extent possible (AFAP) without negatively affecting the benefit-risk ratio. Therefore, even if the risk is acceptable, measures must be implemented to reduce the risk to its lowest level.

The acceptability of the resulting RPN is based on pre-established cut-off values, such as:

RPN <4	Acceptable risk
RPN ≥ 4 but < 15	Tolerable risk
RPN ≥ 15	Unacceptable risk

5.2.4. Risk control

Risk control is the process of deciding whether to reduce or accept identified risks. The approach should follow the risk control hierarchy established in ISO 14971:

- Intrinsically safe design and manufacturing
- Protective measures on the device or in the manufacturing process
- Safety information (e.g., warnings, precautions)

When determining the appropriate control strategy, the following aspects should be considered:

- Predefined risk acceptability criteria
- Effectiveness and feasibility of control measures
- Resources needed to implement and maintain control
- New risks introduced by the control measures themselves

Risk control includes two phases:

a) Risk reduction – Measures implemented to reduce the probability and/or severity of harm. These measures may include design modifications, changes to processes, protective systems, or administrative controls.

b) Risk acceptance – Decision to accept the residual risk without applying further controls, based on a comparison with acceptability criteria or a benefit-risk analysis.

All residual risks must be assessed and, if they are not acceptable according to the defined criteria, a benefit-risk analysis must be performed and documented in accordance with Annex I of the MDR and Clause 8 of ISO 14971.

5.2.5. Residual risk assessment

After implementing risk control measures, all residual risks must be assessed to determine whether they are acceptable and, if not, whether the benefits outweigh the residual risks. Assessing residual risks involves the following steps:

- Estimate the residual risk by reassessing the severity and probability of the damage, taking into account the effectiveness of the controls applied.
- Compare the residual risk with the established predefined risk acceptance criteria.

5.2.6. Risk-benefit analysis

If the residual risk does not meet the acceptability criteria, a documented risk-benefit analysis shall be performed to demonstrate that the medical benefits outweigh the residual risks, as required by:

- ISO 14971:2019, Clause 8
- MDR Annex I, Chapter I, Section 8 and 9

5.2.7. Documentation

All risk management activities shall be properly documented to ensure traceability, accountability, and compliance with ISO 14971:2019, Clause 9, and MDR Regulation 2017/745, Annex I, Section 3.

a) Risk Management File (RMF): A risk management file must be created and maintained for each medical device. This file will provide evidence that the risk management process has been correctly applied throughout the device's life cycle.

The RMF must include, as a minimum:

- The risk management plan (Clause 4.4)
- Records of risk assessment, risk control, residual risk assessment and, where applicable, benefit-risk analysis
- Risk Management Review

b) Risk Management Report: A risk management report must be compiled and approved upon completion of risk management activities. This report must confirm that:

- All identified risks have been assessed and addressed.
- Residual risks are acceptable or have a positive justification of the benefit-risk balance.
- All documentation is complete and up to date
- Risk management is linked to clinical evaluation and post-marketing surveillance (PMS) activities.

Reports will be issued by company personnel. If they are prepared by external companies or subcontracted personnel, they must follow the guidelines of this procedure and be reviewed and approved by the personnel designated by SPIKA TECH in each case.

5.2.8. Information related to the production and post-production phases

Information related to the production phase will be collected during the development process and in process control, where risks related to software development can be identified.

The purpose of collecting device-related information during the post-production phase is to gather information related to usage, product safety, and other relevant aspects. This information can be obtained from various sources, such as:

- as users,
- complaints,
- incident reports, etc.

The information collected is analyzed and processed and may lead to a modification of the list of damages initially identified.

5.2.9. Presentation of results and data review

The risk analysis is reviewed at least once a year or whenever:

- There is an intention to introduce changes to the product, device or process, either by internal initiative or in response to market information
- Post-marketing surveillance data, including user feedback, reveal new risks, harms or hazards
- changes in the state of the art

5.2.10. Risk management applications

As mentioned in section 2. Scope, it applies to the processes carried out to obtain the manufactured products, although there are many interrelated processes, which are detailed below:

- Quality Management System: in the areas of:
  - Documentation
  - Training
  - Quality defects
  - Audits and inspections
  - Change control
- Product design
- Infrastructure: facilities and equipment
  - Design
  - Hygiene
  - Cross contamination
  - Qualification
  - Cleaning
  - Maintenance
- Materials Management: Supplier Evaluation
- Production operations
  - Validation
  - Sampling

6. Distribution of the Procedure

Staff members in the departments listed in Section 3 (Responsibilities and Affected Departments) must be provided with a copy of this procedure. In addition, controlled copy No. 1 will be submitted to the Archives for paper archiving.

Controlled copies to be issued:

Controlled Copy No.	Departments
1	Archive
2	See section no. 3

7. Annexes

NO. / REV.	QUALIFICATION
N/A	N/A

8. Related documentation

Code	QUALIFICATION
MQ-01	Quality Manual
N/A	UNE EN ISO 13485 Medical Devices
N/A	UNE EN ISO 14971 Risk Management of Medical Devices
N/A	ICHQ9

8.1. Training

This version requires that the affected departments indicated in section 3 receive the training indicated below:

Mark with an X	**Type of training**
☒	**Theoretical** (reading and understanding of the procedure)
☐	**Theoretical – Practical** (If you select this option, contact the Responsible Technician)